SYSTEM RULES — ISOLINGO (V0)

SECTION 1 — CORE PURPOSE

V0 is a continuous flashcard-based learning system.

It exists to:
- build vocabulary familiarity
- bind noun + gender into a single reflex
- maximize repetition and exposure

V0 is NOT:
- a testing system
- a progression system
- a performance tracking system

No drills, no challenges, no assessments.

SECTION 2 — CORE LEARNING MODEL (SIMPLIFIED)

Learning is driven by:

1) Pattern Exposure
- repeated encounters with words
- recognition-based interaction

2) Retrieval Practice
- light production (noun recall)
- reinforcement through repetition

The system must:
- show words frequently
- repeat weak items
- mix recognition and production

The system must NOT:
- require rule memorization
- rely on explicit grammar explanation

SECTION 3 — STUDY TYPE BEHAVIOR

User selects one:

1) GENERAL_VOCAB
- includes: nouns, verbs, adjectives, adverbs
- all card types allowed
- noun-specific rules still apply where relevant

2) GENDER
- nouns only
- card pool limited to noun-compatible cards
- heavy emphasis on noun + gender binding

Filtering must occur at session start and persist.

SECTION 4 — ACTIVE VOCAB BANK (V0)

All content must come from the ACTIVE VOCAB BANK.

Each word includes:
- lemma
- English translation
- partOfSpeech
- gender (if noun)

RULES:

- No out-of-bank vocabulary allowed
- Nouns must always include gender metadata
- Words must be reusable across all cards

Minimum dataset:

- ~100 words per language
- Must include sufficient noun diversity

SECTION 5 — CARD SYSTEM (V0)

ONLY the following cards exist:

- CARD 1001 — Matching (Gender Anchoring)
- CARD 1002 — Vocab Relationship Judgment
- CARD 1006 — Noun Production (Direction-Sensitive)

CARD 1002 — SEMANTIC CLARITY RULE

For CARD 1002, the system must ensure that exactly one answer choice is clearly correct.

SYNONYM HANDLING:

- Target-language pairs with identical or near-identical meanings MAY be used
- When used, the correct answer must be:
  "They mean the same thing"

DISTRACTOR RULE:

- For synonym pairs, other answer choices must be clearly incorrect
- Do NOT include distractors that could reasonably apply

AVOID:

- Using synonym pairs where "same category" could also feel correct
- Using pairs where multiple answers depend on interpretation
- Using vague or overlapping semantic relationships

PURPOSE:
Ensure every question has a single, unambiguous correct answer while preserving exposure to semantic relationships.

No additional card types are allowed.

SECTION 6 — CARD ROLE DEFINITIONS

CARD 1001 — Matching
- recognition only

- reinforces gender via color
- low cognitive load

CARD 1002 — Relationship
- recognition + semantic processing
- reinforces vocab discrimination

CARD 1006 — Noun Production (PRIMARY CARD)
- direction-sensitive translation card
- native_to_target remains the primary learning path

German:
- native_to_target requires gender selection
- noun + gender bound

Spanish / French:
- native_to_target requires determiner + noun
- no gender tile
- gender encoded in determiner

target_to_native:
- translation recall only
- no gender output required

SECTION 7 — CARD DISTRIBUTION (CRITICAL)

1006 must be the dominant card.

Target distribution:

GENERAL_VOCAB:
- 1006 → 40–50% of cards (nouns only)
- 1002 → 25–35%
- 1001 → 15–25%

GENDER:
- 1006 → 60–75% (dominant)
- 1001 → 25–40%
- 1002 → optional / low frequency

RULES:

- 1006 must appear frequently and consistently
- no long gaps without production cards
- recognition cards support production, not replace it

SECTION 8 — STUDY SESSION ENGINE

The system runs an infinite loop:

1) Select word(s) (Repetition Engine)
2) Select card type (Card Rotation)
3) Generate card
4) Deliver card
5) Update word stats
6) Repeat

There is no session end.

SECTION 9 — REPETITION ENGINE (SUMMARY)

Each word tracks:
- exposureCount
- errorCount
- frequencyWeight
- lastSeen
- status (NEW / ACTIVE / REINFORCEMENT)

RULES:

- NEW words appear early and frequently
- incorrect words increase in frequency
- correct words decrease gradually
- recently missed words reappear soon

SPACING:

- avoid repeating same word within 2 cards
- prefer spacing within 3–5 cards

SECTION 10 — CARD ROTATION RULES

The system must:

- avoid repeating the same card type consecutively
- alternate between recognition and production
- maintain variation in interaction type

ANTI-AUTOPILOT:

- no same card type >2 times in short window
- no repeated word within 2-card window
- shuffle all answer positions

PRIORITY RULE:

- if production (1006) has not appeared recently,
  it must be prioritized

SECTION 11 — NOUN RESPONSE RULE (CRITICAL)

For noun-production paths where the user is producing the target-language noun:

ABSTRACT RULE (realized differently per language):

A correct answer requires:
1) correct noun
2) correct gender selection

These form ONE combined answer.

VISUAL RULE:

- after selecting gender, typed noun changes to that color
- no duplicate rendering of the noun

INVALID RESPONSES:

- correct noun + wrong gender → incorrect
- wrong noun + correct gender → incorrect

NOUN RESPONSE RULE — LANGUAGE OVERRIDE

German:
- noun + gender tile required
- visual color binding applies

Spanish / French:
- determiner + noun required
- NO gender tile
- NO color binding on input

RATIONALE:
Gender is encoded in determiner choice, not separate selection.

SECTION 12 — LANGUAGE RULES

GERMAN:

- nouns are always color-coded when displayed
- nouns must be capitalized
- no articles shown in vocab cards

SPANISH / FRENCH:

- nouns include article in display
- noun itself is color-coded
- article is NOT color-coded

IDENTICAL-FORM REVERSE-DIRECTION EXCLUSION RULE

For target_to_native cards only:

If the target-language prompt is identical to the English answer
(or so close that the English answer is obvious without real retrieval),
the system must NOT generate that card.

Examples:
- German: Hand → hand

- German: Jeans → jeans
- German: Jobinterview → job interview

RULES:
- This exclusion applies only to target_to_native direction
- It does NOT apply to native_to_target direction
- These words may still appear normally in English → target cards
- The purpose is to avoid unearned, low-value reverse-direction cards

RATIONALE:
- target_to_native should require meaningful recall
- obvious identity matches waste the user's time
- native_to_target remains useful because the learner does not yet know the target form automatically

GENDERED NOUN DISAMBIGUATION RULE

This rule applies to nouns that have distinct masculine and feminine forms referring to people (e.g., German: Kellner / Kellnerin, Spanish: ciudadano / ciudadana).

TARGET → NATIVE (Target Language → English)

- Each target-language noun must be shown individually (never grouped).
- The system must accept natural English equivalents.
- For gender-marked personal nouns, the system must accept both:
  - the natural occupational/person noun
  - a gender-explicit English equivalent when appropriate

Examples:
- Arzt → "doctor", "male doctor"
- Ärztin → "doctor", "female doctor"
- Kellner → "waiter", "server", "male server"
- Kellnerin → "waitress", "server", "female server"
- Schauspieler → "actor", "male actor"
- Schauspielerin → "actress", "female actor", "female actress"
- ciudadano → "citizen", "male citizen"
- ciudadana → "citizen", "female citizen"

RULES:
- Do NOT require artificial translations such as "female doctor" as the only correct answer

- Do accept gender-explicit English answers when they correctly describe the target-language form
- Accept both gender-specific and gender-neutral English where appropriate

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NATIVE → TARGET (English → Target Language)

When a noun has gendered counterparts:

- The system must use a disambiguating English prompt when needed to avoid ambiguity.
- Gender-explicit English prompts are allowed and should be used when they are the clearest way to identify the intended target-language form.

Examples:
- "doctor" may be ambiguous if both Arzt and Ärztin are active
- "male doctor" → Arzt
- "female doctor" → Ärztin
- "waiter" → Kellner
- "waitress" → Kellnerin
- "actor" → Schauspieler when used as masculine/default form
- "actress" → Schauspielerin
- "male citizen" → ciudadano
- "female citizen" → ciudadana

RULES:
- Do NOT use ambiguous prompts like "doctor", "server", "citizen", or "friend" when multiple correct target forms exist, unless multiple answers are explicitly allowed
- The prompt must guide the user toward a single correct target-language answer
- English gendered nouns such as "waitress", "actress", "princess", "mother", "father", "son", "daughter", "brother", and "sister" may be used directly when they naturally identify the intended target form

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SECTION 13 — ANSWER VALIDATION (V0)

Must support:

- correct / incorrect detection
- noun + gender validation (German)
- determiner + noun validation (French / Spanish)
- multiple choice validation

- matching validation

NOT REQUIRED:

- advanced synonym systems
- performance tracking
- analytics

ENGLISH VERB NORMALIZATION RULE

For English verb answers:

The system must treat the following as equivalent:
- bare verb form (e.g., "eat")
- infinitive form with "to" (e.g., "to eat")

RULES:
- Users must NOT be marked incorrect for including or omitting "to"
- Validation must normalize both forms before comparison
- This applies to all cards requiring English verb input (e.g., target_to_native direction)

CANONICAL FORM:
- The stored English value may use either format
- The system must still accept both forms as correct

EXAMPLES:
- Prompt: "essen"
  Accepted answers:
    - "eat"
    - "to eat"

RATIONALE:
- The presence of "to" does not represent a different lexical item in this context
- Rejecting one form creates unnecessary friction and false negatives

DIACRITIC / ACCENT ACCURACY RULE

For all target-language input:

Users must enter the correct spelling, including all required accents and diacritical marks.

SAME-LEMMA ENGLISH ANSWER POOL RULE

For target_to_native cards:

If multiple vocab entries share the same:
- targetLanguage
- normalized lemma

then the system must treat all English glosses from those entries as valid answers for that target-language prompt.

RULES:
- This applies only when the target-language prompt is being translated into English
- The answer pool must include:
  - the current entry's English gloss
  - English glosses from other entries with the same targetLanguage + lemma
  - approved additional accepted English meanings, if present

EXAMPLE:
If the Spanish lemma "música" appears in multiple entries:

- music
- musician
- female musician

Then prompt:
- "la música"

Accepted English answers include:
- "music"
- "musician"
- "female musician"

FEEDBACK RULE:
If the user enters any answer from the same-lemma answer pool:
- mark the answer correct
- show normal correct feedback
- also show the other accepted meanings

Suggested text:
"Also correct: [other accepted meanings]"

RATIONALE:
Some target-language forms legitimately map to multiple English meanings.
The user should not be penalized for entering a linguistically correct answer tied to the same lemma.

COMMA-SEPARATED ENGLISH ANSWERS RULE

When the "english" field contains multiple comma-separated values,
these values represent interchangeable accepted answers.

RULES:
- The user is NOT expected to type all values
- The user may enter ANY ONE of the listed values
- Each value must be treated as a fully correct answer

EXAMPLE:
english: "free, available"

Accepted user answers:
- "free" ✔
- "available" ✔

Rejected:
- "free, available" (not expected as a combined response)

FEEDBACK RULE:
- After a correct answer, the system must display:
  "Also correct: [other accepted values]"

EXAMPLE:
User enters: "free"
System displays:
- "Correct"
- "Also correct: available"

RELATION TO SAME-LEMMA RULE:
- Comma-separated values are treated as a local answer pool for a single entry
- SAME-LEMMA pools are treated as a cross-entry answer pool
- Both rules may apply simultaneously

RULES:
- Missing or incorrect accents must be marked incorrect
- Anglicized substitutions are NOT accepted
  Examples:
    - "ueber" is NOT accepted for "über"
    - "cafe" is NOT accepted for "café"
- This applies to all supported languages

RATIONALE:

- Accents and diacritics are part of the correct spelling of the word
- Accepting approximations reduces learning precision and creates bad habits

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SECTION 14 — FLIP BEHAVIOR

All cards include Flip.

On Flip:
- correct answer is shown
- input is disabled
- Next button appears

RULES:

- no penalty for flipping
- no restriction on use

SECTION 15 — GLOBAL UX RULES

All cards must include:

Top left:
- Exit Study

Top right:
- Home

Bottom:
- Flip (always visible)
- Submit or interaction
- Next (after resolution)

RULE:

- no dead-end states
- user can always continue or exit

SECTION 16 — HARD EXCLUSIONS

Do NOT include:

- drills
- challenges
- assessments
- scoring
- streaks
- progress tracking
- difficulty levels
- competency systems beyond vocab/gender

SECTION 17 — FINAL RULE

If a feature is not defined here:

→ DO NOT BUILD IT